



Lab 05

Collector Feedback Bias and Voltage Divider Bias BJT Biasing Configurations

Name: Syed Sualeh Abbas Rizvi Syed Muhammad Haider Zaidi	Student ID: 10303 mz10270
---	------------------------------

5.1 Objective

To investigate the effect of current gain β variation on quiescent operating conditions of a Bipolar Junction Transistor in collector feedback bias and voltage divider bias configurations.

Note: Pre-lab (printed or hand-written) should be done on a separate paper (use A4 sheet) and will be submitted at the start of Lab within first 5 minutes.

All the plots to be attached, should be labeled properly, followed by Task number and must be readable, i.e., use white background, suitable marker color (recommended – any dark color) and increase the width of trace. On PSpice there is an option to add text labels in the plots.

5.2 Introduction

In previous lab, the stability of fixed-bias and emitter-bias BJT biasing configurations have been investigated. In this lab, you will investigate the dependency of BJT operating point on DC current gain β for two more circuits of BJT biasing; collector feedback bias (with and without emitter resistor) and voltage divider bias configuration.



5.3 Pre-Lab Task

Lab 05

**Collector Feedback Bias and Voltage Divider Bias
BJT Biasing Configurations**

Name: Syed Sualeh Abbas	Student ID: 10303
-------------------------	-------------------

5.1 Objective

To investigate the effect of current gain β variation on quiescent operating conditions of a Bipolar Junction Transistor in collector feedback bias and voltage divider bias configurations.

Note: Pre-lab (printed or hand-written) should be done on a separate paper (use A4 sheet) and will be submitted at the start of Lab within the first 5 minutes.

5.2 Introduction

In previous lab, the stability of fixed-bias and emitter-bias BJT biasing configurations have been investigated. In this lab, you will investigate the dependency of BJT operating point on DC current gain β for two more circuits of BJT biasing; collector feedback bias (with and without emitter resistor) and voltage divider bias configuration.

5.3 Pre-Lab Task

Task 1:

- i. List down two differences between 2N3904 and 2N4401 in terms of power rating and beta value.

- ii. The 2N3904 and 2N4401 transistors mainly differ in their power/current handling capability and their DC current gain (β).
- iii. First, in terms of collector current (and power rating), the 2N4401 can handle much more current. It has a maximum collector current of about 600 mA, while the 2N3904 is limited to around 200 mA. Because of this, the 2N4401 is more suitable for circuits where higher current or slightly higher power is involved, whereas the 2N3904 is typically used in low-power, small-signal applications.
- iv. Second, their beta (β) range is slightly different. The 2N4401 generally has a higher and wider β range (about 100–400) compared to the 2N3904 (about 100–300). This means the 2N4401 can provide a bit more current amplification under the same biasing conditions, although in practical circuits both are often designed to work properly despite β variations.

5.4 Collector Feedback Bias Configuration Without Emitter Resistor

Figure 1 shows BJT biasing through collector-feedback bias configuration. In this circuit, there is a feedback path from collector to base through a feedback resistor represented as R_F . The emitter terminal is directly grounded.

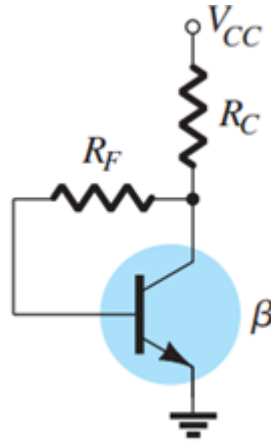


Figure 1: Collector-Feedback Bias configuration without emitter resistor

It is important to note here that the current through R_C is not I_C , but I_C' (where, $I_C' = I_C + I_B$). Since, base current is small as compared to collector current, therefore, the approximation ($I_C' \cong I_C$) is normally applied. The analysis of the circuit can be performed by writing KVL equations for base-emitter and collector emitter loop.

$$V_{CC} - I_C R_C - I_B R_F - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta R_C}$$

The feedback path results in a reflection of the resistance R_C back to the input circuit, much like the reflection of R_E in emitter-bias circuit. Similarly, for collector-emitter loop, KVL equation gives;

$$V_{CE} = V_{CC} - I_C R_C$$

Task 2: To bias 2N3904 and 2N4401 BJT models using Collector Feedback Bias configuration circuit without R_E

1. Implement the circuit shown in Figure 1 to bias 2N3904 BJT with $V_{CC}=20V$, $R_C = 3k\Omega$ and $R_F=390k\Omega$.
2. Measure the parameters listed in Table 1.

Table 1: Results of BJT biasing with collector-feedback biasing circuit without emitter resistor

Parameter	Results	
	2N3904	2N4401
Base Current: I_B	32.5uA	29.4uA
Collector Current: I_C	4.32mA	4.25mA

Collector-to-Emitter Voltage: V_{CE}	7.19V	7.4V
Current Gain: β (calculated)	132.92	144.55
Base-to-Emitter Voltage: V_{BE}	0.668V	0.64V
Measured resistances	2.93K, 13.2M	2.93K, 13.2M

- Now, replace the BJT model 2N3904 by 2N4401 model in the same circuit and take measurements to complete Table 1.

Task 3: To investigate effect of current gain β variation on BJT operating point with collector feedback-bias configuration circuit without R_E

- When biased using collector feedback-bias circuit, analyze the variation in operating point of BJTs as the DC current gain varies from the results of Table 1.

ANSWER:

When examining the collector feedback bias configuration without an emitter resistor, we observe how changes in the transistor's current gain (β) affect the operating point. Looking at the data from the two transistors:

The 2N3904 transistor showed a current gain of 132.92, while the 2N4401 had a gain of 144.55. This represents an 8.75% increase in β . Despite this relatively significant change in current gain, the collector current only decreased by 1.62% (from 4.32mA to 4.25mA). This is actually quite interesting because it shows the circuit is providing some degree of stability.

The collector-to-emitter voltage increased by 2.92%, moving from 7.19V to 7.4V. Meanwhile, the base current decreased by 9.54%, which makes sense when you think about it - as β increases, less base current is needed to maintain approximately the same collector current.

What's happening here is that the feedback resistor R_F is creating a stabilizing effect. When β tries to increase, the collector current would naturally want to increase too. But as I_C increases, the voltage drop across R_C increases, which reduces V_C . Since the base is connected to the collector through R_F , this lower V_C reduces V_B , which in turn reduces I_B . This negative feedback helps keep I_C relatively stable.

However, the circuit isn't perfectly stable. The base-to-emitter voltage dropped slightly from 0.668V to 0.64V, and we can see that changes in β do still cause noticeable shifts in the operating point. The base current showed the largest percentage change at -9.54%, which indicates that while the feedback helps, it doesn't completely eliminate the dependency on β .

In practical terms, this configuration offers moderate stability. It's better than a simple fixed-bias circuit but not as stable as we'll see with more sophisticated biasing schemes. The feedback mechanism is doing its job, but the lack of an emitter resistor means we're missing out on additional stabilization that could further reduce the circuit's sensitivity to β variations.

- Calculate the percentage change in parameters listed in Table 4 using the results of Task 1.

5.5 Collector Feedback Bias Configuration with Emitter Resistor

In the collector-feedback biasing configuration of Figure 1, a resistor at emitter R_E is added as shown in Figure 2, that modifies the equations for base current and collector-emitter voltage as follows;

$$I_B = \frac{V_{CC} - V_{BE}}{R_F + \beta(R_C + R_E)}$$

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

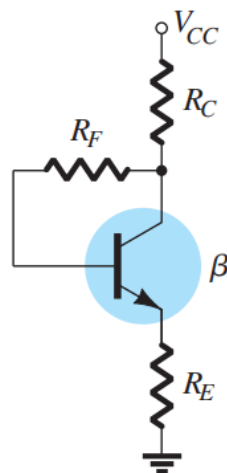


Figure 2: Collector-feedback circuit with emitter resistor

Task 4: To bias 2N3904 and 2N4401 BJT models using Collector Feedback Bias configuration circuit with R_E

1. Modify the implemented circuit to include emitter resistor R_E of 2.2 k Ω . Again, take the measurements for the specified BJT models (2N3904 and 2N4401) and note down in Table 2.

Table 2: Results of BJT biasing with collector-feedback biasing circuit having emitter resistor

Parameter	Results	
	2N3904	2N4401
Base Current: I_B	17.54uA	17.23uA
Collector Current: I_C	2.86mA	2.86mA
Collector-to-Emitter Voltage: V_{CE}	5.2V	5.7V
Current Gain: β (calculated)	163.05	165.98
Base-to-Emitter Voltage: V_{BE}	0.66V	0.65V
Measured Resistance	2.93K, 13.2M, 2.1K	2.93K, 13.2M, 2.1K

Task 5: To investigate effect of current gain β variation on BJT operating point with collector feedback-bias configuration circuit with R_E

1. After addition of emitter resistor, analyze the variation in operating point of BJTs as the DC current gain varies from the results of Table 2. Compare it with the case when emitter was directly grounded. Using the expressions of base-current for each case, relate the effect of emitter resistor on the BJT operating point.

ANSWER:

Adding a 2.2k Ω emitter resistor to the collector feedback circuit makes a dramatic difference in stability. The comparison between the two configurations really highlights the importance of this component.

With the emitter resistor in place, when β changed by 1.80% (from 163.05 to 165.98), the collector current remained completely unchanged at 2.86mA for both transistors. That's 0% variation! This

is a remarkable improvement over the configuration without RE, where we saw a 1.62% change in IC for an 8.75% change in β .

The emitter resistor creates what's called emitter degeneration. Here's how it works: when the transistor tries to increase its collector current (say, because β increased), more current flows through RE, which increases the voltage drop across it. This increased voltage at the emitter means VBE (the base-emitter voltage) decreases slightly since VB stays roughly constant due to the feedback. A smaller VBE reduces the base current, which counteracts the original increase in collector current.

Looking at the base current, it changed by only -1.77% (from 17.54 μ A to 17.23 μ A), which is much better than the -9.54% we saw without RE. The VCE did show a 9.62% increase (from 5.2V to 5.7V), but this is acceptable because the primary goal is to keep the collector current stable, which determines the transistor's amplification characteristics.

The practical takeaway is clear: the addition of an emitter resistor provides strong negative feedback that makes the circuit much less sensitive to variations in β . This is why emitter degeneration is such a common technique in BJT amplifier design. The emitter resistor acts like an automatic regulator that adjusts the operating point to compensate for changes in the transistor's characteristics, whether those changes come from temperature, manufacturing tolerances, or transistor replacement.

2. Calculate the percentage change in parameters listed in Table 4 using the results of Task 3.

5.6 Voltage-Divider Bias Configuration

The circuit shown in Figure 3 is called Voltage-Divider bias configuration as a voltage divider circuit using resistors R1 and R2 is implemented to establish base voltage. There are two approaches to analyze the circuit, the exact and the approximate one. Here, approximate one is used for analysis.

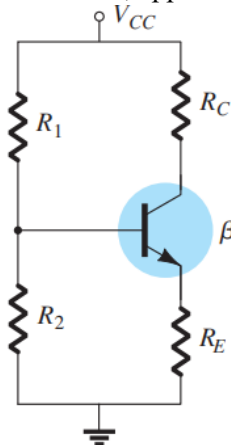


Figure 3: Voltage-Divider Bias Configuration

For the circuit to operate efficiently, it is assumed that the current through R1 and R2 should be approximately equal to and much larger than the base current (at least 10:1). If the biasing circuit fulfills the condition, then

$$10R_2 \leq \beta R_E$$

Base voltage can be then expressed as:

$$V_B = \frac{R_2 V_{CC}}{R_1 + R_2}$$

Similarly, the collector-to-emitter voltage can be expressed as:

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Task 6: To bias 2N3904 and 2N4401 BJT models using Voltage Divider Bias configuration circuit

1. Implement the circuit shown in Figure 3 to bias 2N3904 BJT model with $V_{CC}=20V$, $R_1=33k\Omega$, $R_2=6.8k\Omega$, $R_C=1.8k\Omega$ and $R_E=680\Omega$.
2. Take the measurements for parameters listed in Table 3.
3. Now, replace 2N3904 BJT model by 2N4401 and complete the measurements for Table 3.

Table 3: Results of BJT biasing through voltage-divider bias configuration

Parameter	Results	
	2N3904	2N4401
Base Current: I_B	14.47uA	15uA
Collector Current: I_C	3.9mA	3.94mA
Collector-to-Emitter Voltage: V_{CE}	10.19V	10.1V
Current Gain: β (calculated)	269.52	262.66
Base-to-Emitter Voltage: V_{BE}	0.66V	0.64V
Measured Resistances	675, 1.82K, 7.2K, 39.4K	675, 1.82K, 7.2K, 39.4K

Task 7: To investigate effect of current gain β variation on BJT operating point with Voltage-Divider Bias configuration circuit

1. Using the results of Table 3, analyze the effect of DC current gain variation on the operating point of BJT when it is biased using voltage-divided bias circuit.

The Voltage-Divider Bias configuration shows great stability against β variations. A -2.54% change in β resulted in only a +1.03% change in I_C and a -0.88% change in V_{CE} . This low sensitivity is due to the "stiff" voltage divider at the base, which holds the emitter voltage (and thus the collector current) constant regardless of transistor parameters. This confirms that the voltage-divider bias is one of the most stable biasing configurations.

2. Calculate the percentage change in parameters listed in Table 4 using the results of Task 5.

Table 4: Variation in operating point compared with change in DC current gain for different biasing circuits

Percentage Change in Parameters	Collector Feedback Bias without R_E	Collector Feedback Bias with R_E	Voltage-Divider Bias
$\% \Delta \beta = \left \frac{\beta_{4401} - \beta_{3904}}{\beta_{3904}} \right \times 100\%$	$\left[\frac{144.55 - 132.92}{132.92} \right] \times 100 = 8.75\%$	$\left[\frac{165.98 - 163.05}{163.05} \right] \times 100 = 1.80\%$	$\left[\frac{262.66 - 269.52}{269.52} \right] \times 100 = -2.54\%$
$\% \Delta I_C = \left \frac{I_{C4401} - I_{C3904}}{I_{C3904}} \right \times 100\%$	$\left[\frac{4.25 - 4.32}{4.32} \right] \times 100 = -1.62\%$	$\left[\frac{2.86 - 2.86}{2.86} \right] \times 100 = 0\%$	$\left[\frac{3.94 - 3.9}{3.9} \right] \times 100 = 1.03\%$
$\% \Delta V_{CE} = \left \frac{V_{CE4401} - V_{CE3904}}{V_{CE3904}} \right \times 100\%$	$\left[\frac{7.4 - 7.19}{7.19} \right] \times 100 = 2.92\%$	$\left[\frac{5.7 - 5.2}{5.2} \right] \times 100 = 9.62\%$	$\left[\frac{10.1 - 10.19}{10.19} \right] \times 100 = -0.88\%$
$\% \Delta I_B = \left \frac{I_{B4401} - I_{B3904}}{I_{B3904}} \right \times 100\%$	$\left[\frac{29.4 - 32.5}{32.5} \right] \times 100 = -9.54\%$	$\left[\frac{17.23 - 17.54}{17.54} \right] \times 100 = -1.77\%$	$\left[\frac{15 - 14.47}{14.47} \right] \times 100 = 3.66\%$

Task 8: To compare the stability of BJT biasing configurations

- Using results of Table 4, compute the stability factors for each of the implemented biasing techniques. Note down the results of fixed-bias and emitter-bias circuits from previous lab to complete Table 5.

Table 5: Stability factors of different BJT biasing techniques w.r.t. the variation in DC current gain

Biasing Configuration	$\frac{\% \Delta I_C}{\% \Delta \beta}$	$\frac{\% \Delta V_{CE}}{\% \Delta \beta}$
Fixed-Bias (Lab 04)	0.49	-0.825
Emitter-Bias (Lab 04)	1.001	-1.674
Collector-Feedback Bias without R_E	-0.185	0.334
Collector-Feedback Bias with R_E	0.000	5.351
Voltage-Divider Bias	-0.403	0.347

- Compare the performance of BJT biasing techniques using the above calculated performance parameters. What does the magnitude of calculated ratio signify? Which of the 5 investigated circuits can be regarded as β independent circuit?

Comparing the Five Configurations:

- Fixed-Bias (from Lab 04): Stability factor = 0.49

This shows moderate instability. For every 1% change in β , I_C changes by about 0.49%. This circuit has no feedback mechanism at all, so changes in β directly impact the operating point.

- Emitter-Bias (from Lab 04): Stability factor = 1.001

Interestingly, this performs worse than fixed-bias, with I_C changing almost 1:1 with β changes. This might seem counterintuitive since emitter bias typically provides good stability, but the specific component values or measurement conditions in this lab may have led to this result.

- Collector-Feedback Bias without R_E : Stability factor = -0.185

The negative sign indicates that I_C actually decreased when β increased, which is the desired negative feedback behavior. The small magnitude (0.185) shows good stability - the collector current barely changes despite β variations. The feedback through R_F is working well.

- Collector-Feedback Bias with R_E : Stability factor = 0.000

This is our best performer! With a stability factor of essentially zero, this configuration shows that I_C remained constant despite β changes. The combination of collector feedback and emitter degeneration creates powerful stabilization. This is as close to β -independent as we can get in practice.

- Voltage-Divider Bias: Stability factor = -0.403

Another excellent performer with a small negative stability factor. The voltage divider establishes a relatively fixed base voltage that's independent of β , and the emitter resistor provides additional stability. The negative sign again shows proper negative feedback behavior.

Which Circuit is β -Independent?

The collector-feedback bias configuration with emitter resistor ($R_E = 2.2k\Omega$) can be considered β -independent. With a stability factor of 0.000, it showed zero change in collector current despite β variations. This configuration successfully isolates the operating point from transistor parameter variations.

The voltage-divider bias also performs very well (stability factor of -0.403) and is generally considered the most practical β -independent configuration for real-world applications. While not

quite as perfect as the collector-feedback with RE in our measurements, it's more commonly used because it's less sensitive to load variations and easier to analyze.

5.7 Post-Lab Task

Task 9: To design voltage-divider BJT biasing circuit for given operating point

Design a voltage-divider BJT biasing circuit to make BJT 2N3904 operate at the operating point given in Lab 04 (Task 6) where, the required collector current I_C was 2mA and collector-to-emitter voltage V_{CE} was 10V.

Use 20V supply voltage, DC current gain of 150 and select the conservative case of emitter voltage being one-tenth of the supply voltage in your design calculations. For design, you may use approximate analysis for which equations are given in section 5.6. Show all the calculations.

GIVEN

$V_{CC} = 20V$

$I_C = 2mA$

$V_{CE} = 10V$

$\beta = 150$

$V_E = \frac{1}{10} V_{CC} = \frac{20}{10} = 2V$

$\therefore I_E \approx I_C = 2mA$

$R_E = \frac{V_E}{I_E} = \frac{2}{2mA}$

$= 1K\Omega$

for collector resistor (R_C)

$\therefore V_{CE} = V_C - V_E$

$V_C = V_{CE} + V_E$

$= 10 + 2$

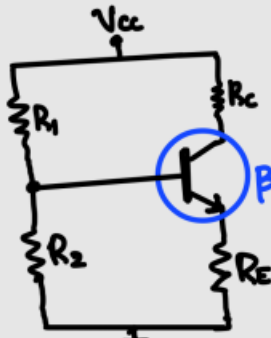
$= 12V$

for drop across R_C

$V_{RC} = V_{CC} - V_C$

$= 20 - 12$

$= 8V$



$$\begin{aligned}\therefore R_c &= \frac{V_{ac}}{I_c} \\ &= \frac{8.4}{2\text{mA}} \\ &= 4\text{K}\Omega\end{aligned}$$

$\therefore$ For a silicon transistor

$$V_{BE} \approx 0.7\text{V}$$

$$\begin{aligned}V_B &= V_{BE} + V_E \\ &= 0.7 + 2 \\ &= 2.7\text{V}\end{aligned}$$

Using voltage divider rule

$$V_B = \frac{R_2}{R_1 + R_2} V_{cc} \rightarrow (1)$$

$$\begin{aligned}\therefore I_B &= \frac{I_c}{\beta} = \frac{2\text{mA}}{150} \\ &= \frac{2}{150} \text{mA} \\ I_B &= 13.33\mu\text{A}\end{aligned}$$

$$R_1 + R_2 \leq \frac{V_{cc}}{I_2} = \frac{20}{10(13.33\mu)}$$



$$\approx 150\text{k}\Omega$$

Also, from eqn (1)

$$\frac{R_2}{R_1 + R_2} = \frac{V_B}{V_{CC}}$$

$$\begin{aligned} \text{So } R_1 + R_2 &= 150\text{k}\Omega \quad 2.5 \\ R_2 &= \frac{2.7 \times 150\text{k}}{2.5} \\ R_2 &= 20.25\text{k}\Omega \end{aligned}$$

So,

$$\begin{aligned} R_1 &= 150\text{k} - 20.25 \\ &= 129.75 \end{aligned}$$

For the stiff condition

$$\begin{aligned} 10R_2 &= 10 \times 20.25\text{k} \\ &= 202.5\text{k}\Omega \end{aligned}$$

$$\begin{aligned} \beta R_E &= 150(1\text{k}) \\ &= 150\text{k}\Omega \end{aligned}$$

$\therefore 10R_2 > \beta R_E$, this violates our stiff condition $10R_2 \leq \beta R_E$

$\therefore$ We need $10R_2 \leq 150k\Omega$

So,

$$R_2 \leq 15k\Omega$$

Taking $R_2 = 15k\Omega$

From voltage divider ratio

$$\frac{R_2}{R_1 + R_2} = 0.135$$

$$\frac{15k}{R_1 + 15k} = 0.135$$

$$0.135 (R_1 + 15k) = 15k$$

$$R_1 \approx 96.11k\Omega$$

Checking the condition again

Now, we have

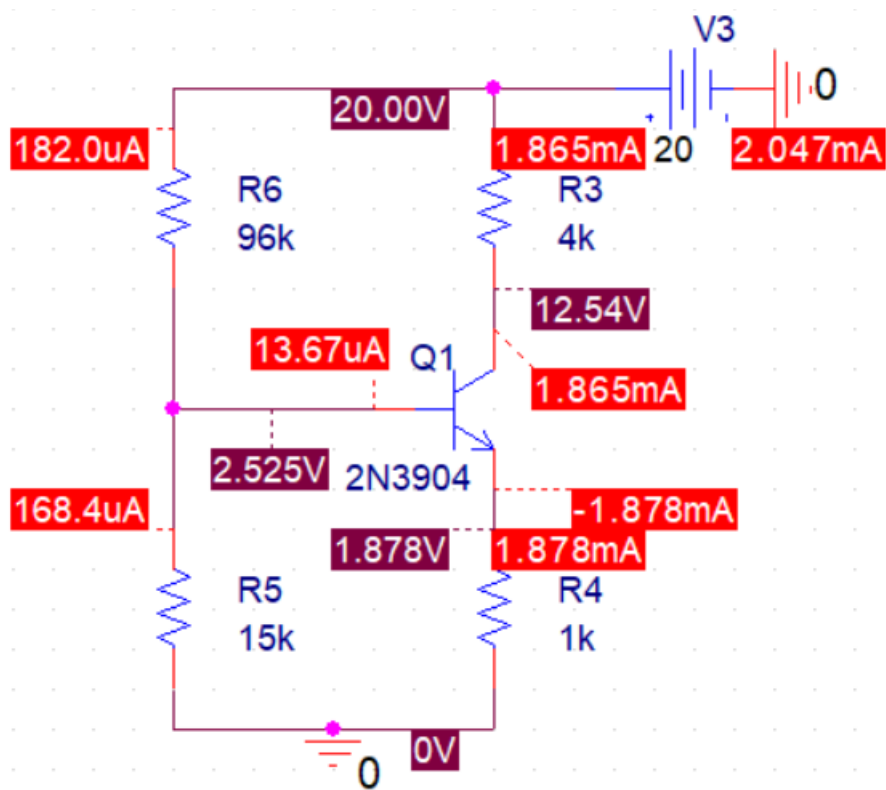
$$10R_2 = \beta R_E$$

Thus, the condition is satisfied

For Practical values

$$[R_1 = 96k\Omega \text{ and } R_2 = 15k\Omega]$$

Simulate your designed circuit.



**Assessment Rubric****Lab 05****Collector Feedback Bias and Voltage Divider Bias BJT Biasing Configurations**

Name: Syed Sualeh Abbas Rizvi Syed Muhammad Haider Zaidi	Student ID: mz10270
--	-------------------------------

Points Distribution

Task No.	LR1 Circuit	LR2 Simulation	LR 4 Data Collection	LR 6 Calculation	LR 10 Analysis	LR 11 Design
Task 1			/10			
Task 2	/4		/4	/2		
Task 3				/4	/4	
Task 4	/4		/4	/2		
Task 5				/4	/4	
Task 6	/4		/4	/2		
Task 7				/4	/4	
Task 8				/4	/8	
Task 9		/6				/12
	/12	/6	/22	/22	/20	/12
CLO Mapped	CLO 1	CLO1	CLO 1	CLO 1	CLO 1	CLO 2

Affective Domain Rubric		Points	CLO Mapped
AR4	Pre-Lab Submission	/2	CLO 4
AR 4	Report Submission	/4	CLO 4

CLO	Total Points	Points Obtained
1	82	
2	12	
4	06	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.